The spatial diffusion of simple and complex technologies

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Sunset

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The spatial diffusion of simple and complex technologies

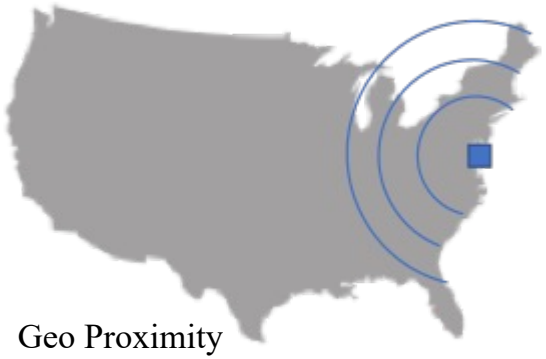
Diffusion

Complexity

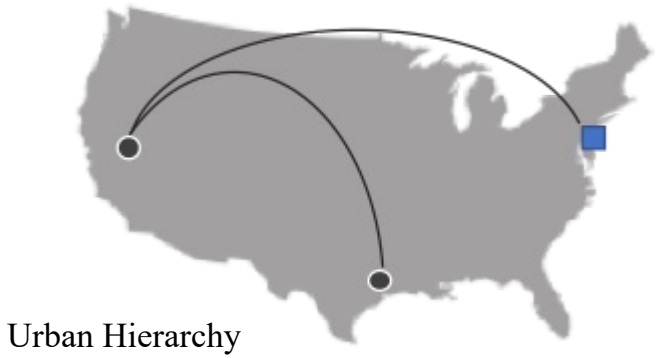
Theory 1: Diffusion of technologies

1. The exploration of diffusion is **not conclusive** in Economic Geography

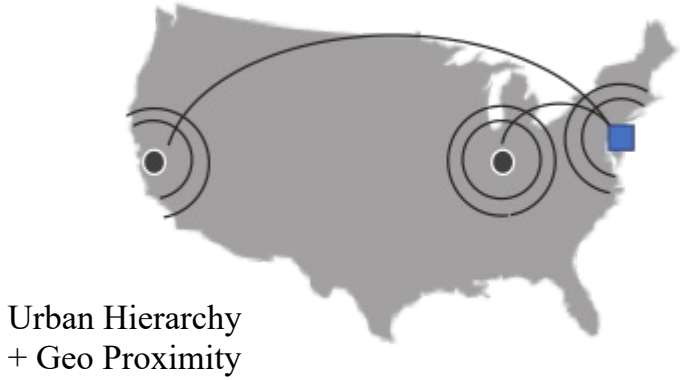
Contagious diffusion



Leap-like diffusion



Hierarchical diffusion



(Hägerstrand, 1967)

2. Lately, studies show that **different factors might matter in diffusion of technologies**

- Geographic distance (*Hägerstrand, 1967, Balland and Rigby, 2017)*
- Technological distance *(Boschma and Frenken, 2011)*
- Social distance (*Sorenson et al., 2006)* etc.

3. In this research, we are focused on investigating how **geographic distance** might influence diffusion

Geographic Distances

Dist_origin

Dist_nearest_adopter

Novelty

– Isolating the impact of **dist_origin** in the diffusion process – regions that invent the technology!

One could debate…

Dist_origin would not matter!	Dist_origin matters a lot!
○ **Dist_nearest_adopter** is what matters! ○ You can easily learn from nearby adopters!	○ Learning mostly happens from the region where the technology was invented ○ Adopters - ☑ adopt tech ☑ recombine tech ? diffusing tech ○ Potential Bais? ○ Inventors want to **diffuse a lot** ○ Potential Adopters might want to **adopt from inventors**

Theory 2: Complexity

WHY do we expect the relevance of distance for technological adoption to vary as a function of complexity?

One could debate…

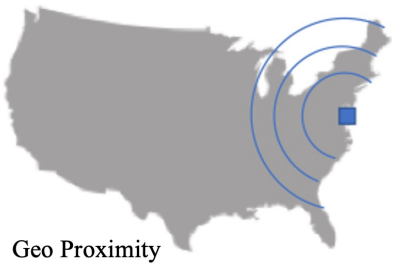
Closer the better!	Distance is irrelevant!
○ **Complex technologies are non-ubiquitous** and their knowledge is **tacit** in large parts (*Balland and Rigby, 2017)* ○ **Complex technologies are harder to adopt!** Adopting complex technologies can be time and resource-intensive (*Cohen and Levinthal, 1990)* – so closer location might help	Interested regions can adopt complex technologies from any distance given: ○ **Monopolistic Rents** (*Broekel, 2019)* ○ **Competitive Advantage** (*Hidalgo & Hausmann, 2009)* and ○ **Economic Growth** (*Broekel & Klarl, 2025; Mewes & Broekel, 2022; Sbardella et al., 2018; Boltho et al.,2018; Pugliese et al., 2017)*

*Does more complex technology **diffuse differently** than less complex technology?*

Hypotheses

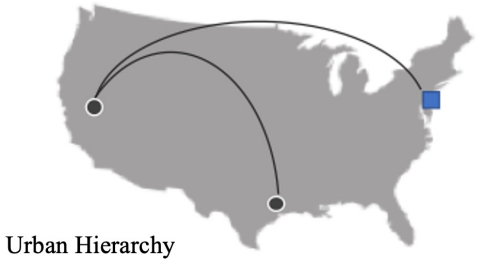
H1. More complex technology diffuses in a **contagious fashion**

Contagious diffusion



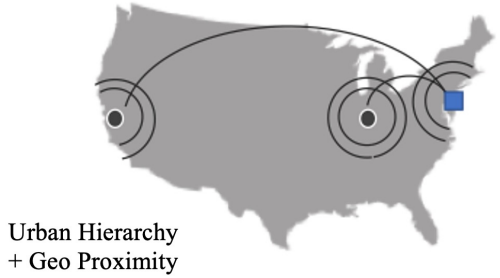
H2. More complex technology diffuses in a **leap-like fashion**

Leap-like diffusion



H3. More complex technology diffuses in a **hierarchical fashion**

Hierarchical diffusion



Data

US patents from HistPat and USPTO (1836 to 2021)

New technology in this research is a combination of two 4-digit CPCs

A01B_A61P

WHY?
1. The re-combinatorial nature of innovation (Mewes, 2019)
2. Allows us to assign more appropriate complexity values

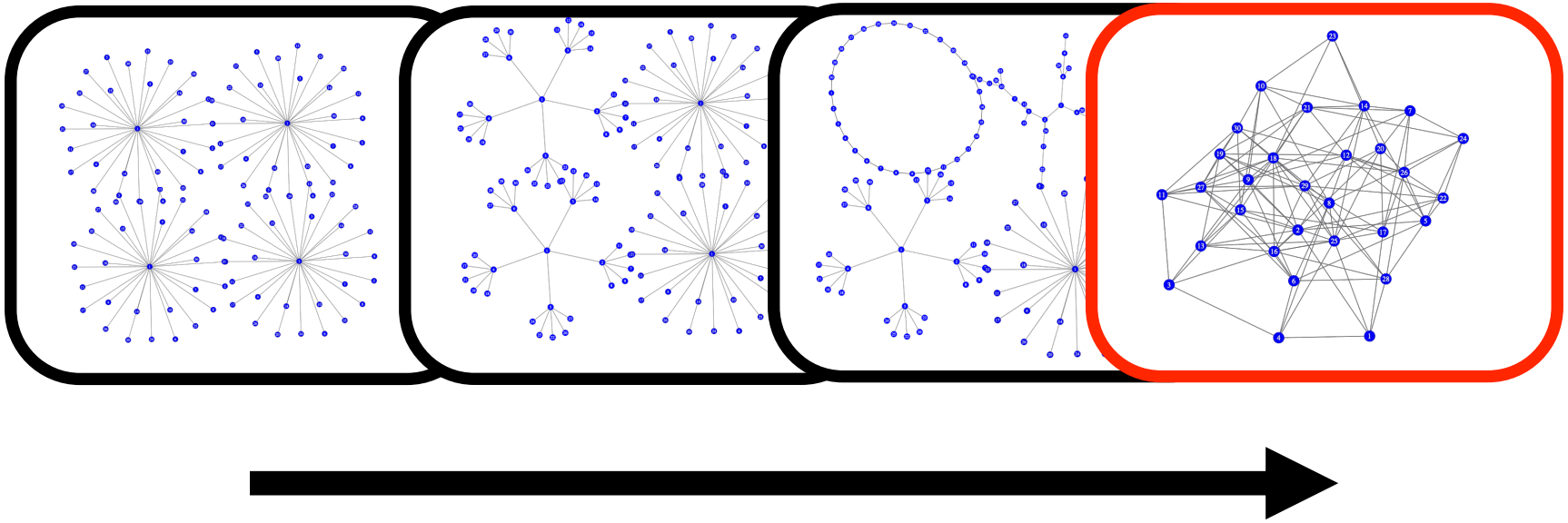
Total technologies (that diffused to minimum 50 regions): 5,752
Total MSAs*: 363
Analysis at 5-years time-period starting at 1836

**Metropolitan Statistical Area*

Approximating Complexity

Empirical measure of **structural diversity** for patent data *(Broekel, 2019)*

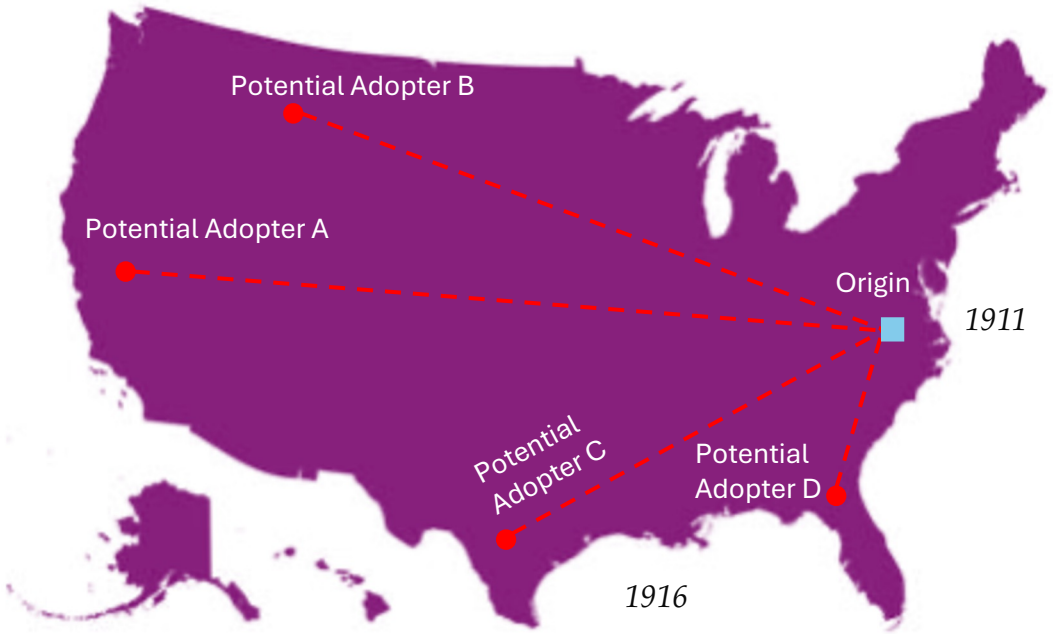
Topological diversity of how technological sub-components (10-digit CPC) classes are combined



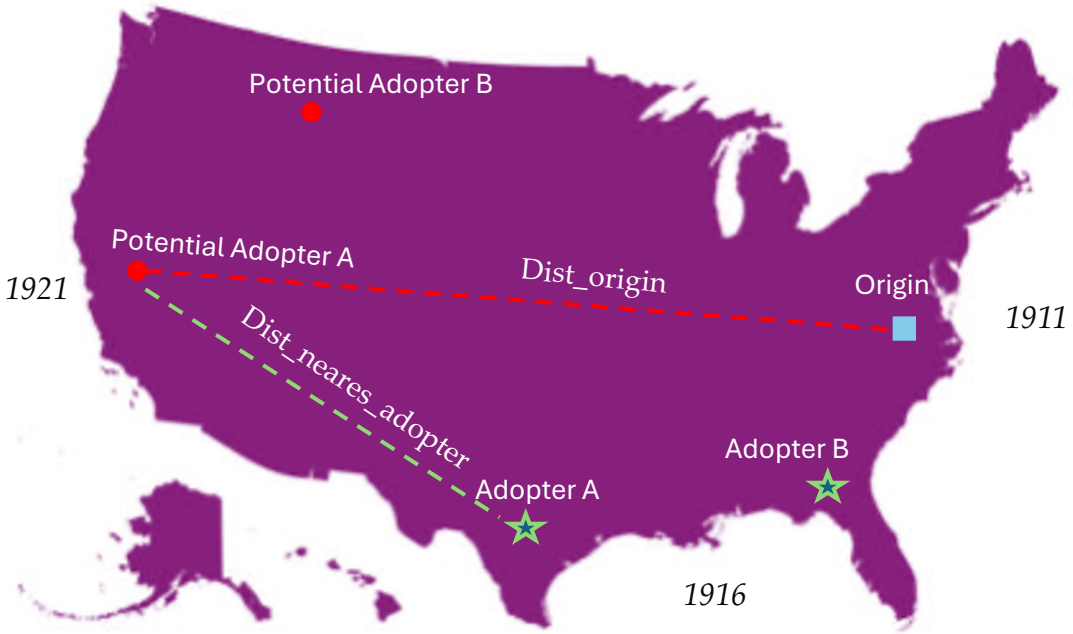
Structural diversity

Approximating Distances

Dist_origin

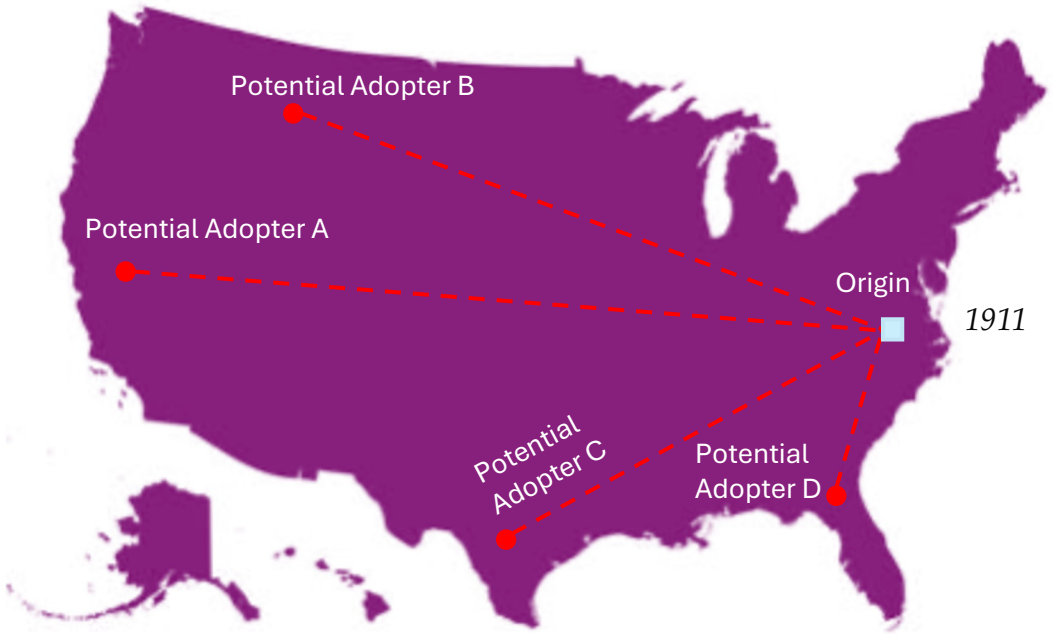


Dist_nearest_adopter



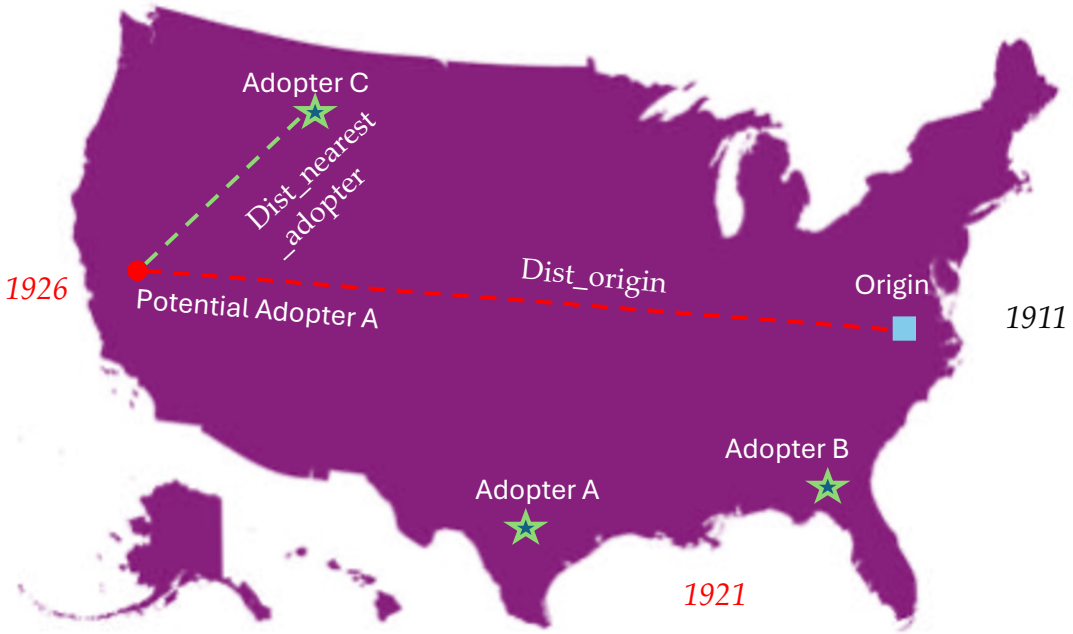
Approximating Distances

Dist_origin



Always Static

Dist_nearest_adopter



Extremely Dynamic

Model

Adoption ~ *dist_origin + dist_nearest_adopter + complexity +*

*dist_origin*complexity + dist_nearest_adopter*complexity +*

Controls

population + n_patents +

n_patents_cpc1 + n_patents_cpc2 +

social_proximity +

relatedness_cpc1 + relatedness_cpc2 +

region_tech_similarity_cpc1 + region_tech_similarity_cpc2 +

region (MSA) FE + time (duration) FE

Sample Dataframe

Adopter Regions	Year	Adoption?	Distance Origin	Distance Adopter	Technological Proximity	Social Proximity	Complexity
Region XYZ	1891	0	50 km	150 km	.32	.25	.5
Region XYZ	1896	0	50 km	150 km	.51	.26	.6
Region XYZ	1901	0	50 km	125 km	.77	.26	.6
Region XYZ	1911	0	50 km	125 km	.79	.30	.6
Region XYZ	1916	0	50 km	50 km	.81	.40	.6
Region XYZ	1921	0	50 km	50 km	.89	.40	.7
Region XYZ	1926	1	50 km	50 km	.89	. 40	.7
Region ABC	1891	0	100 km	100 km	.76	.34	.5
Region ABC	1896	0	100 km	100 km	.77	.43	.6
Region ABC	1901	0	100 km	17 km	.79	.54	.6
Region ABC	1911	0	100 km	3 km	.81	.55	.6
Region ABC	1916	0	100 km	3 km	.89	.67	.6
Region ABC	1921	1	100 km	3 km	.91	.78	.7
⋮	⋮	⋮	~ 35m rows	⋮	⋮	⋮	⋮

Estimation Strategy

Can rely on ordinary OLS or Logit models because a lot of 0 in adoption

Hence, we mostly use a **Cox proportional hazards model**

- Designed for time-to-event data
- Naturally handles many non-adopters (right-censoring)

Main advantage:

- The model estimates the hazard of adoption
- Interpreted as the instantaneous probability of adopting at time t, given not yet adopted

Prelim result!

Technology diffusion models

	Logit FE	Cox (strata)	Cox (complex)
Distance to origin	1.002	1.008***	1.157***
	(0.002)	(0.002)	(0.007)
Distance to nearest adopter	0.894***	0.895***	0.924***
	(0.003)	(0.002)	(0.007)
Technological complexity			1.084***
			(0.001)
Distance to origin × complexity			0.986***
			(0.001)
Distance to nearest adopter × complexity			0.996***
			(0.001)
Population	1.318***	1.801***	1.660***
	(0.009)	(0.011)	(0.011)
Patents	0.397***	0.741***	0.792***
	(0.003)	(0.004)	(0.005)
CPC variety (level 1)	2.098***	1.514***	1.521***
	(0.003)	(0.001)	(0.001)
CPC variety (level 2)	2.029***	1.418***	1.421***
	(0.003)	(0.001)	(0.001)
Social proximity	0.996	1.130***	0.999
	(0.003)	(0.003)	(0.003)
Relatedness (max)	1.543***	1.798***	1.852***
	(0.011)	(0.011)	(0.011)
Relatedness (min)	1.645***	1.350***	1.328***
	(0.010)	(0.006)	(0.006)
MSA-tech similarity (max)	1.089***	1.165***	1.136***
	(0.006)	(0.006)	(0.006)
MSA-tech similarity (min)	1.188***	1.260***	1.248***
	(0.006)	(0.005)	(0.005)
Observations	34990256	34990256	34990256

Entries are exponentiated coefficients. Logit FE reports odds ratios; Cox models report hazard ratios. Standard errors in parentheses.

Estimation Strategy

Split technologies into **20 complexity bins/ranks**

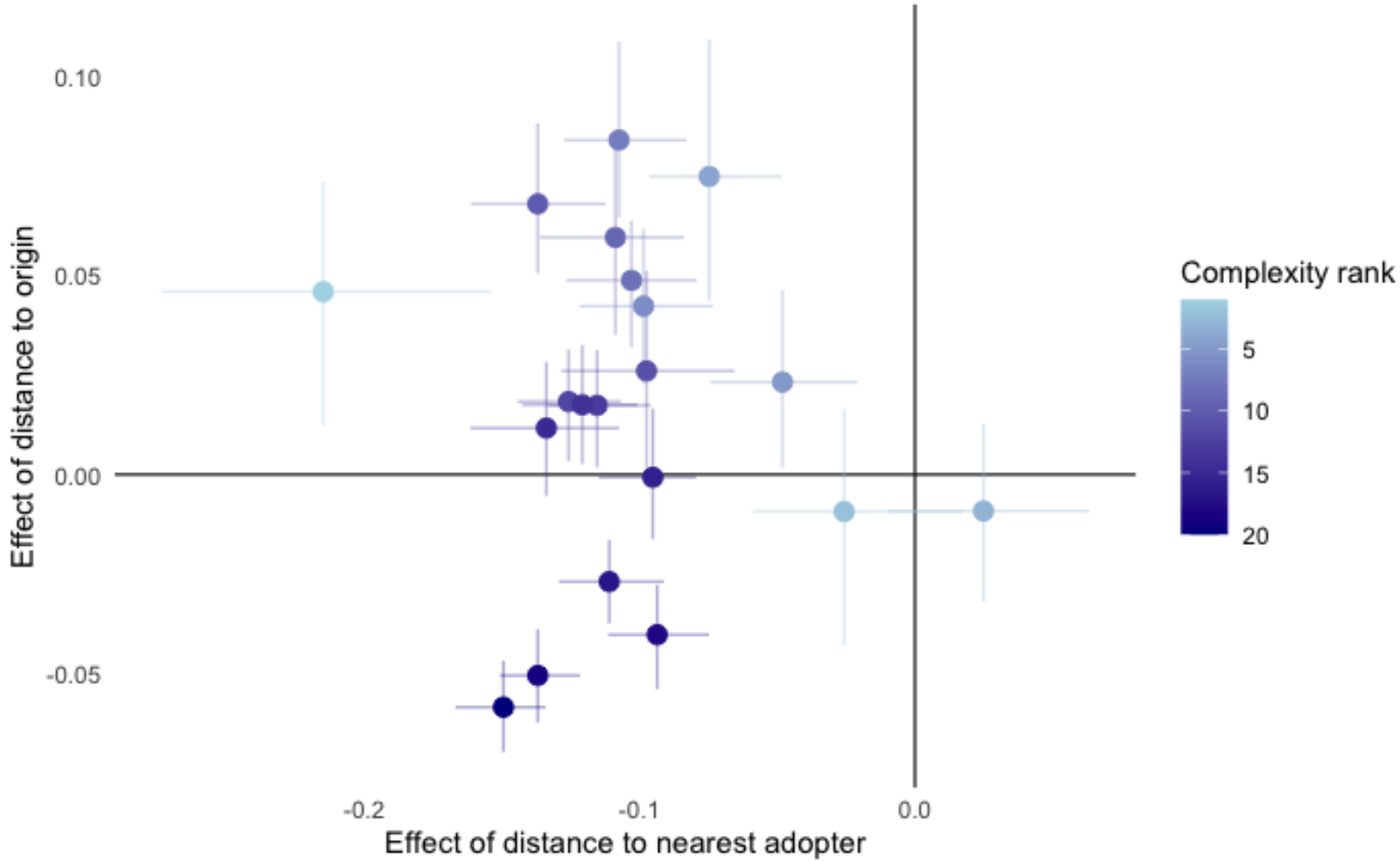
- To allow for flexible, non-linear effects and avoid imposing a functional form

Used **Bootstrap resampling**

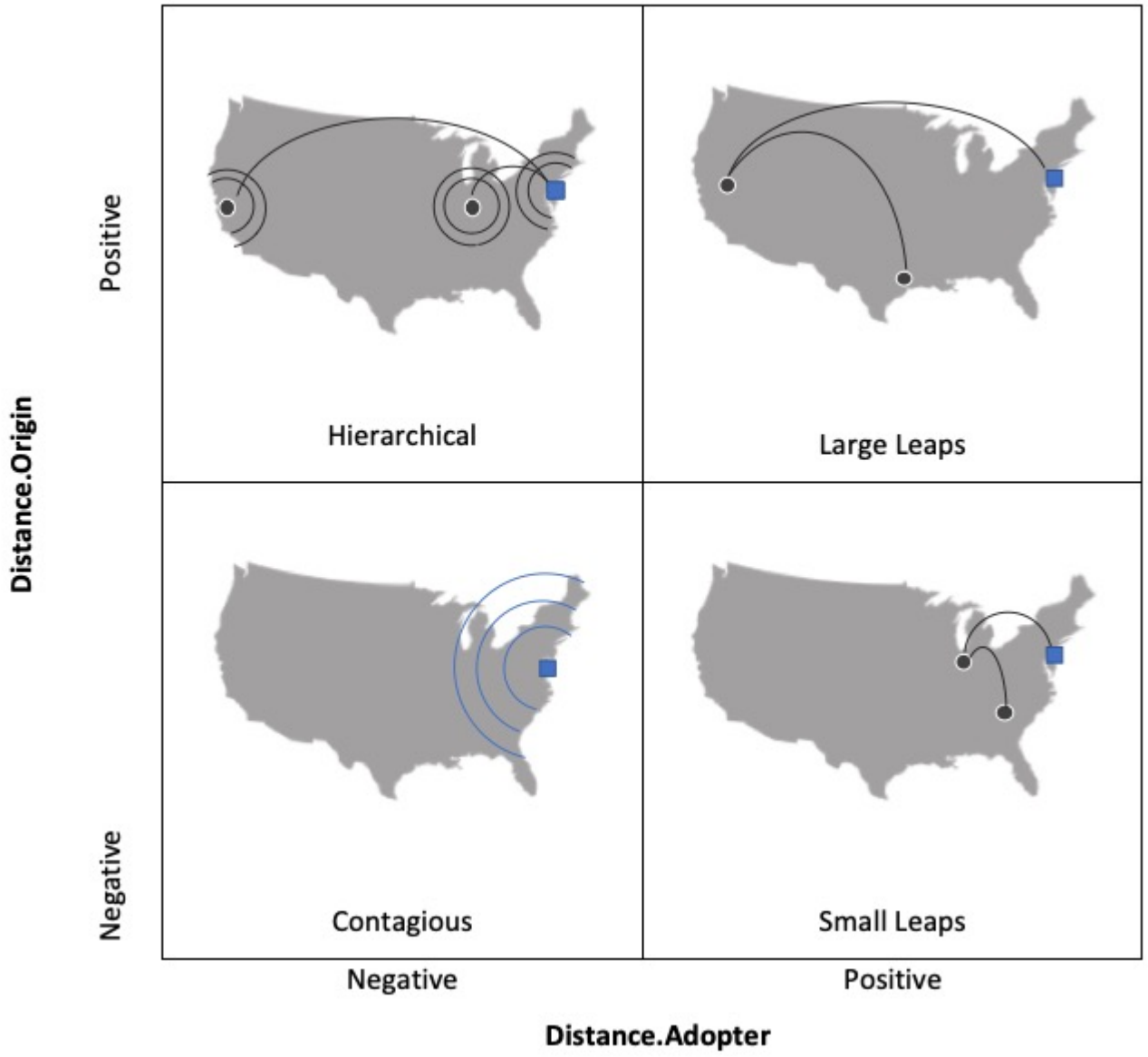
- To obtain robust confidence intervals
- To ensure that results are not driven by sample-specific variation

Prelim result!

Confidence intervals based on 300 bootstrap samples



Diffusion pattern



REACH OUT!

Thank You!

Would love some feedback!

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Approximating Technological Relatedness

1. How close is each MSAs' tech portfolio to the focal technology?

2. How close is each MSAs' tech portfolio to tech portfolio of the adopter?

Approximating Technological Relatedness

1. How close is each MSAs' tech portfolio to the focal technology?

For each time period

1. Built a **cpc_co-occurrence** matrix

2. Converted co-occurrence into **cpc_relatedness_matrix**

3. Then computed **relatedness_density**, based on:
   - **region_cpc_matrix** (what regions already do) and
   - **cpc_relatedness_matrix** (how technologies are connected)

4. Assigned the maximum relatedness_density as the **technological_relatedness** value

(EconGeo Package)

Approximating Technological Relatedness

2. How close is each MSAs' tech portfolio to tech portfolio of the adopter?

For each time period

1. Built a **cpc_co-occurrence** matrix
2. Built a **msa_cpc_mat** identifying the share of patents in each technology within each region
3. Constructed **technology-weighted regional profiles** by weighting each region's CPC portfolio using relatedness to focal technologies
4. Computed **cosine similarity** between the technology-weighted profiles of potential adopters and all adopting regions
5. Assigned the maximum cosine similarity as the **msa_tech_similarity** value

(EconGeo Package)